

TRAN LY MINH HOANG

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github.com/TLMHoang-AI

PROFESSIONAL SUMMARY

AI Engineering student with hands-on experience in **computer vision, speech/audio processing, and deep learning** using Python and PyTorch. Experienced in researching, developing, evaluating, and optimizing AI solutions through **model design, data preprocessing, feature engineering, and systematic experimentation**, with additional experience building end-to-end AI pipelines.

TECHNICAL SKILLS

Programming:	Python, SQL
AI / Deep Learning:	PyTorch, TensorFlow, Keras, Scikit-learn, Computer Vision, Object Detection, Semantic Segmentation, Model Optimization
Data & Signal Processing:	OpenCV, NumPy, Pandas, librosa, Image Processing, Audio Processing, Feature Engineering
Backend & Data:	FastAPI, Celery, Redis, PostgreSQL, MySQL, SQLite
Infrastructure:	Docker, Docker Compose, AWS, Microsoft Azure database services, Linux
Development & Research Tools:	Git, GitHub, Jupyter, Google Colab, Kaggle, Jira, LaTeX
Languages:	Vietnamese (Native), English (IELTS 6.5)

EDUCATION

FPT University

Major: Artificial Intelligence

Nov 2023 – Present

EXPERIENCE

FPT Software

AI Engineer & Python Developer Intern

Internship

Sep 2025 – Dec 2025

- Developed components for an internal **software modernization platform** that transforms legacy COBOL, JCL, and Assembly systems into modern Java and Python applications while analyzing source-code components and dependencies.
- Automated a **Python-based data processing pipeline**, transforming source files and structured JSON outputs into language-specific Excel reports for engineering analysis and validation.
- Supported application development and testing using **Microsoft Azure database services, AWS cloud environments, and Jira-based engineering workflows**.

RELEVANT PROJECTS

U-DeepCrack: Road Crack Segmentation

[GitHub](#)

Team Leader / AI Engineer / Solution Researcher | Team Size: 3

- Led research and development of a **PyTorch-based computer vision system** for road-crack segmentation, extending an existing architecture with improved multi-scale feature reconstruction for detecting thin and difficult crack patterns.
- Designed and evaluated **23 controlled training configurations**, comparing loss functions and adaptive optimization strategies to improve model accuracy and cross-dataset generalization.

- Evaluated the optimized model on an independent dataset, achieving **41.26% Dice** and **30.35% IoU** on OmniCrack30K after training and validation on UDTIRI-Crack.

Bee Swarming Detection from Acoustic Signals

[GitHub](#)

Team Leader / AI Engineer / Solution Researcher | Team Size: 3

- Led development of an end-to-end **speech/audio processing and machine-learning pipeline**, extracting MFCC, STFT, and Chroma representations followed by feature selection, dimensionality reduction, model training, and evaluation.
- Systematically evaluated and optimized SVM, Random Forest, Extra Trees, and ensemble approaches across two audio datasets, achieving **99.22% test accuracy** with the strongest feature configuration.

Text2Sign: Vietnamese Text-to-Sign Language System

[GitHub](#)

AI Engineer / Solution Researcher | Team Size: 3

- Built components of an end-to-end **language-to-motion AI pipeline** that transforms Vietnamese text into sign-language animation through text processing, dictionary resolution, motion retrieval, human-motion processing, and 3D avatar animation.
- Integrated the AI pipeline with **FastAPI, Celery, Redis, PostgreSQL, and Docker Compose**, connecting AI processing with API services, background tasks, persistent storage, and containerized application components.

CONFERENCE PAPER & AWARD

Impact of DICOM Preprocessing Strategies on Chest X-ray Abnormality Detection

APWeb-WAIM 2026

AI Engineer / Solution Researcher | Team Size: 4

Best Paper Award – Special Track

- Researched and evaluated **computer vision preprocessing strategies** for YOLOv8, Faster R-CNN, and RT-DETR, systematically comparing multiple image representations for medical abnormality detection.
- Conducted model evaluation on **6,471 curated VinDr-CXR images across 14 abnormality classes**; the strongest RT-DETR configuration achieved **35.58% mAP@0.5**.

COMPETITIONS

Research Connect 2026

Finalist – Top 12 Teams

Team Leader | Road Crack Segmentation

2026

- Led a comparative research study of three computer vision architectures, coordinating model development, optimization, performance analysis, and cross-dataset evaluation.

Research Connect 2025

Finalist – Top 12 Teams

Team Leader | LeafNet – Plant Leaf Disease Classification

2025

- Led research and evaluation of handcrafted-feature, deep-feature, CNN, and Transformer-based approaches for image classification; the strongest DenseNet201 model achieved **95.70% test accuracy** and **95.55% weighted F1-score**. [GitHub](#)